

Distribution of Y_i in SLR

$$\begin{aligned}\mathbb{E}[Y_i] &= \mathbb{E}[\beta_0 + \beta_1 x_i + \epsilon_i] = \beta_0 + \beta_1 x_i + \mathbb{E}[\epsilon_i] \quad \text{Note: } \epsilon \sim N(0, \sigma^2) \\ &= \beta_0 + \beta_1 x_i\end{aligned}$$

$$\text{Var}(Y_i) = \text{Var}(\beta_0 + \beta_1 x_i + \epsilon_i) = \text{Var}(\epsilon_i) = \sigma^2$$

Since ϵ_i has a normal distribution, so does Y_i . Thus, the distribution of Y_i is

$$Y_i \sim N(\beta_0 + \beta_1 x_i, \sigma^2)$$

Note that the Y_i 's are independent but NOT identically distributed since the means are changing through x_i

MLEs of β_0 and β_1 in SLR

$$\begin{aligned}f(y_i|\beta) &= (2\pi\sigma^2)^{-1/2} \exp\left\{-\frac{1}{2}\left(\frac{y_i - (\beta_0 + \beta_1 x_i)}{\sigma}\right)^2\right\} \\ L(\beta) &= \prod_{i=1}^n f(y_i|\beta) = (2\pi\sigma^2)^{-n/2} \exp\left\{-\sum_{i=1}^n \frac{1}{2}\left(\frac{y_i - (\beta_0 + \beta_1 x_i)}{\sigma}\right)^2\right\} \\ \mathcal{L}(\beta) &= -\frac{n}{2} \log 2\pi\sigma^2 - \sum_{i=1}^n \frac{1}{2}\left(\frac{y_i - (\beta_0 + \beta_1 x_i)}{\sigma}\right)^2\end{aligned}$$

$$\begin{aligned}\frac{\partial \mathcal{L}(\beta)}{\partial \beta_0} &= 0 - \left[\frac{1}{2\sigma^2} \left(2 \sum (y_i - \beta_0 - \beta_1 x_i) \right) (-1) \right] \\ &= \frac{1}{\sigma^2} \sum (y_i - \beta_0 - \beta_1 x_i) \\ \implies 0 &= \sum y_i - n\beta_0 - \beta_1 \sum x_i \\ \implies n\beta_0 &= n\bar{y} - \beta_1 \sum x_i \\ \implies \hat{\beta}_0 &= \bar{y} - \hat{\beta}_1 \bar{x}\end{aligned}$$

$$\begin{aligned}\frac{\partial \mathcal{L}(\beta)}{\partial \beta_1} &= 0 - \left[\frac{1}{2\sigma^2} \left(2 \sum (y_i - \beta_0 - \beta_1 x_i) \right) (-x_i) \right] \\ &= \frac{1}{\sigma^2} \sum (y_i x_i - \beta_0 x_i - \beta_1 x_i^2) \\ \implies 0 &= \sum y_i x_i - \beta_0 \sum x_i - \beta_1 \sum x_i^2 \\ \implies 0 &= \sum y_i x_i - (\bar{y} - \beta_1 \bar{x}) \sum x_i - \beta_1 \sum x_i^2 \\ \implies 0 &= \sum y_i x_i - \bar{y} \sum x_i + \beta_1 \bar{x} \sum x_i - \beta_1 \sum x_i^2 \\ \implies \bar{y} n \bar{x} - \sum y_i x_i &= \beta_1 \left(n \bar{x}^2 - \sum x_i^2 \right) \quad \text{Multiply by } (-1) \\ \implies \sum y_i x_i - n \bar{x} \bar{y} &= \beta_1 \left(\sum x_i^2 - n \bar{x}^2 \right) \\ \implies \hat{\beta}_1 &= \frac{\sum x_i y_i - n \bar{x} \bar{y}}{\sum x_i^2 - n \bar{x}^2} = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sum (x_i - \bar{x})^2} = \frac{S_{XY}}{S_{XX}}\end{aligned}$$

Estimates of β_0 and β_1 by minimizing Q

$$Q = \sum_{i=1}^n \left(y_i - (\hat{\beta}_0 + \hat{\beta}_1 x_i) \right)^2$$

$$\begin{aligned} \frac{\partial Q}{\partial \beta_0} &= 2 \sum (y_i - \beta_0 - \beta_1 x_i) (-1) = -2 \sum (y_i - \beta_0 - \beta_1 x_i) \\ &\implies 0 = \sum y_i - n\beta_0 - \beta_1 \sum x_i \\ &\text{Same as MLE so} \\ &\implies \hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x} \end{aligned}$$

$$\begin{aligned} \frac{\partial Q}{\partial \beta_1} &= 2 \sum (y_i - \beta_0 - \beta_1 x_i) (-x_i) = -2 \sum (y_i - \beta_0 - \beta_1 x_i) (x_i) \\ &\implies 0 = \sum x_i y_i - \beta_0 \sum x_i - \beta_1 \sum x_i^2 \\ &\implies 0 = \sum y_i x_i - (\bar{y} - \beta_1 \bar{x}) \sum x_i - \beta_1 \sum x_i^2 \\ &\text{Same as MLE so} \\ &\implies \hat{\beta}_1 = \frac{S_{XY}}{S_{XX}} \end{aligned}$$

Sampling Distribution of $\hat{\beta}_1$

Recall,

$$\hat{\beta}_1 = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sum (x_i - \bar{x})^2} = \frac{\sum (x_i - \bar{x}) y_i - \sum (x_i - \bar{x}) \bar{y}}{\sum (x_i - \bar{x})^2} = \frac{\sum (x_i - \bar{x}) y_i}{S_{XX}} = \sum c_i y_i \quad \text{where } c_i = \frac{x_i - \bar{x}}{S_{XX}}$$

Note that the y_i 's are normally distributed. Thus $\hat{\beta}_1$ is a linear combination of normal random variables and thus is also normal.

$$\begin{aligned} \mathbb{E}[\hat{\beta}_1] &= \mathbb{E}\left[\sum c_i y_i\right] = \sum c_i \mathbb{E}[y_i] \\ &= \sum c_i (\beta_0 + \beta_1 x_i) \\ &= \beta_0 \sum \frac{x_i - \bar{x}}{S_{XX}} + \beta_1 \sum \frac{(x_i - \bar{x}) x_i}{S_{XX}} \quad \text{Note that } \sum (x_i - \bar{x}) = 0 \\ &= \beta_1 \sum \frac{(x_i - \bar{x}) x_i}{S_{XX}} \\ &= \frac{\beta_1}{S_{XX}} \sum (x_i - \bar{x}) x_i \end{aligned}$$

Lets work with just $\sum (x_i - \bar{x}) x_i$

$$\begin{aligned} \sum (x_i - \bar{x}) x_i &= \sum (x_i^2 - \bar{x} x_i) = \sum (x_i^2 - 2x_i \bar{x} + \bar{x}^2) + x_i \bar{x} - \bar{x}^2 \\ &= \sum (x_i - \bar{x})^2 + \sum (x_i \bar{x} - \bar{x}^2) \\ &= \sum (x_i - \bar{x})^2 + \bar{x} \sum x_i - n \bar{x}^2 \\ &= \sum (x_i - \bar{x})^2 + n \bar{x}^2 - n \bar{x}^2 \\ &= \sum (x_i - \bar{x})^2 \\ &= S_{XX} \end{aligned}$$

So then we have that

$$\mathbb{E} \left[\hat{\beta}_1 \right] = \frac{\beta_1}{S_{XX}} \sum (x_i - \bar{x}) x_i = \frac{\beta_1 S_{XX}}{S_{XX}} = \beta_1$$

$$\begin{aligned} \text{Var} \left(\hat{\beta}_1 \right) &= \text{Var} \left(\sum c_i y_i \right) = \sum c_i^2 \text{Var} (y_i) \quad \text{Based on assumption that } y_i \text{'s are independent} \\ &= \sigma^2 \sum c_i^2 \\ &= \sigma^2 \sum \left(\frac{x_i - \bar{x}}{S_{XX}} \right)^2 \\ &= \frac{\sigma^2}{S_{XX}^2} \sum (x_i - \bar{x})^2 \\ &= \frac{\sigma^2 S_{XX}}{S_{XX}^2} \\ &= \frac{\sigma^2}{S_{XX}} \end{aligned}$$

So then we have that,

$$\hat{\beta}_1 \sim N \left(\beta_1, \frac{\sigma^2}{S_{XX}} \right)$$

Sampling Distribution of $\hat{\beta}_0$

Note that $\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$ is a linear combination of normal random variables ($\hat{\beta}_1$ and y_i 's) and thus is also normal.

$$\begin{aligned} \mathbb{E} \left[\hat{\beta}_0 \right] &= \mathbb{E} \left[\bar{y} - \hat{\beta}_1 \bar{x} \right] = \mathbb{E} [\bar{y}] - \bar{x} \mathbb{E} [\hat{\beta}_1] \\ &= \mathbb{E} \left[\frac{\sum y_i}{n} \right] - \bar{x} \beta_1 \\ &= \frac{1}{n} \sum \mathbb{E} [y_i] - \beta_1 \bar{x} \\ &= \frac{1}{n} \sum \mathbb{E} [\beta_0 + \beta_1 x_i + \epsilon_i] - \beta_1 \bar{x} \\ &= \frac{1}{n} \sum (\beta_0 + \beta_1 x_i) - \beta_1 \bar{x} \\ &= \frac{n\beta_0}{n} + \frac{\beta_1 \sum x_i}{n} - \beta_1 \bar{x} \\ &= \beta_0 + \beta_1 \bar{x} - \beta_1 \bar{x} \\ &= \beta_0 \end{aligned}$$

$$\text{Var} \left(\hat{\beta}_0 \right) = \text{Var} \left(\bar{y} - \hat{\beta}_1 \bar{x} \right) = \text{Var} (\bar{y}) + \bar{x}^2 \text{Var} \left(\hat{\beta}_1 \right) - 2\text{Cov} \left(\bar{y}, \hat{\beta}_1 \right)$$

Let's solve this in pieces. First, we will start with $\text{Var} (\bar{y})$.

$$\text{Var} (\bar{y}) = \text{Var} \left(\frac{\sum y_i}{n} \right) = \frac{1}{n^2} \sum \text{Var} (y_i) = \frac{1}{n^2} \sum \text{Var} (\beta_0 + \beta_1 x_i + \epsilon_i) = \frac{1}{n^2} \sum \sigma^2 = \frac{n\sigma^2}{n^2} = \frac{\sigma^2}{n}$$

From the derivation of the distribution of $\hat{\beta}_1$, we know that

$$\text{Var} \left(\hat{\beta}_1 \right) = \frac{\sigma^2}{S_{XX}} \implies \bar{x}^2 \text{Var} \left(\hat{\beta}_1 \right) = \frac{\sigma^2 \bar{x}^2}{S_{XX}}$$

Finally, we must solve $2\text{Cov}(\bar{y}, \hat{\beta}_1)$. Recall from the derivation of the distribution of $\hat{\beta}_1$ that

$$\hat{\beta}_1 = \sum c_i y_i \quad \text{where} \quad c_i = \frac{x_i - \bar{x}}{S_{XX}}$$

We can use this to solve for the desired covariance.

$$\begin{aligned} \text{Cov}(\bar{y}, \hat{\beta}_1) &= \text{Cov}\left(\frac{\sum y_i}{n}, \sum c_i y_i\right) = \frac{1}{n} \sum c_i \text{Cov}(y_i, y_i) \\ &= \frac{1}{n} \sum c_i \text{Var}(y_i) \\ &= \frac{1}{n} \sum c_i \sigma^2 \\ &= \frac{\sigma^2}{n} \sum c_i \\ &= \frac{\sigma^2}{n} \sum \frac{x_i - \bar{x}}{S_{XX}} \\ &= \frac{\sigma^2}{n S_{XX}} \sum (x_i - \bar{x}) \\ &= 0 \end{aligned}$$

Put all the pieces together to get that

$$\text{Var}(\hat{\beta}_0) = \text{Var}(\bar{y}) + \bar{x}^2 \text{Var}(\hat{\beta}_1) - 2\text{Cov}(\bar{y}, \hat{\beta}_1) = \frac{\sigma^2}{n} + \frac{\sigma^2 \bar{x}^2}{S_{XX}} - 0 = \sigma^2 \left[\frac{1}{n} + \frac{\bar{x}^2}{S_{XX}} \right]$$

So then we have that

$$\hat{\beta}_0 \sim N\left(\beta_0, \sigma^2 \left[\frac{1}{n} + \frac{\bar{x}^2}{S_{XX}} \right]\right)$$

Sampling Distribution of $\hat{\mu}^*$

Note that $\mu^* = \mathbb{E}[Y^*] = \mathbb{E}[\beta_0 + \beta_1 x^*] = \beta_0 + \beta_1 x^*$. Also note that $\hat{\mu}^* = \hat{\beta}_0 + \hat{\beta}_1 x^*$ is a linear combination of normal random variables, thus $\hat{\mu}^*$ is also normal.

$$\mathbb{E}[\hat{\mu}^*] = \mathbb{E}[\hat{\beta}_0 + \hat{\beta}_1 x^*] = \mathbb{E}[\hat{\beta}_0] + x^* \mathbb{E}[\hat{\beta}_1] = \beta_0 + \beta_1 x^*$$

$$\begin{aligned} \text{Var}(\hat{\mu}^*) &= \text{Var}(\hat{\beta}_0 + \hat{\beta}_1 x^*) = \text{Var}(\hat{\beta}_0) + (x^*)^2 \text{Var}(\hat{\beta}_1) + 2x^* \text{Cov}(\hat{\beta}_0, \hat{\beta}_1) \\ &= \sigma^2 \left[\frac{1}{n} + \frac{\bar{x}^2}{S_{XX}} \right] + (x^*)^2 \frac{\sigma^2}{S_{XX}} + 2x^* \text{Cov}(\hat{\beta}_0, \hat{\beta}_1) \end{aligned}$$

We need to solve for $\text{Cov}(\hat{\beta}_0, \hat{\beta}_1)$

$$\text{Cov}(\hat{\beta}_0, \hat{\beta}_1) = \text{Cov}(\bar{y} - \hat{\beta}_1 \bar{x}, \hat{\beta}_1) = \text{Cov}(\bar{y}, \hat{\beta}_1) - \bar{x} \text{Cov}(\hat{\beta}_1, \hat{\beta}_1)$$

Recall from derivation for distribution of $\hat{\beta}_0$ that $\text{Cov}(\bar{y}, \hat{\beta}_1) = 0$

$$\begin{aligned} &= 0 - \bar{x} \text{Var}(\hat{\beta}_1) \\ &= -\frac{\bar{x} \sigma^2}{S_{XX}} \end{aligned}$$

Thus we get that

$$\begin{aligned}
\text{Var}(\hat{\mu}^*) &= \sigma^2 \left[\frac{1}{n} + \frac{\bar{x}^2}{S_{XX}} \right] + (x^*)^2 \frac{\sigma^2}{S_{XX}} + 2x^* \text{Cov}(\hat{\beta}_0, \hat{\beta}_1) \\
&= \sigma^2 \left[\frac{1}{n} + \frac{\bar{x}^2}{S_{XX}} \right] + \frac{(x^*)^2 \sigma^2}{S_{XX}} - \frac{2x^* \bar{x} \sigma^2}{S_{XX}} \\
&= \sigma^2 \left[\frac{1}{n} + \frac{\bar{x}^2}{S_{XX}} + \frac{(x^*)^2}{S_{XX}} - \frac{2x^* \bar{x}}{S_{XX}} \right] \\
&= \sigma^2 \left[\frac{1}{n} + \frac{1}{S_{XX}} \left((x^*)^2 - 2x^* \bar{x} + \bar{x}^2 \right) \right] \\
&= \sigma^2 \left[\frac{1}{n} + \frac{(x^* - \bar{x})^2}{S_{XX}} \right]
\end{aligned}$$

So then we have that

$$\hat{\mu}^* \sim N \left(\beta_0 + \beta_1 x^*, \sigma^2 \left[\frac{1}{n} + \frac{(x^* - \bar{x})^2}{S_{XX}} \right] \right)$$

Sampling Distribution of $\hat{Y}^*$

Note that $\hat{Y}^* = \hat{\beta}_0 + \hat{\beta}_1 x^*$. Also note that $\hat{Y}^* = \hat{\beta}_0 + \hat{\beta}_1 x^*$ is a linear combination of normal random variables, thus $\hat{Y}^*$ is also normal.

Consider $y^* - \hat{y}^*$.

$$\mathbb{E}[y^* - \hat{y}^*] = \mathbb{E}[\beta_0 + \beta_1 x^* + \epsilon^*] - \mathbb{E}[\hat{\beta}_0 + \hat{\beta}_1 x^*] = (\beta_0 + \beta_1 x^*) - (\beta_0 + \beta_1 x^*) = 0$$

$$\text{Var}(y^* - \hat{y}^*) = \text{Var}(y^*) + \text{Var}(\hat{y}^*) + 2\text{Cov}(y^*, \hat{y}^*)$$

Note that $2\text{Cov}(y^*, \hat{y}^*) = 0$ since y^* is some future value for a given x^* . Thus $\hat{y}^*$ is independent of it.

Also note that $\text{Var}(\hat{y}^*) = \text{Var}(\hat{\mu}^*)$ since $\hat{y}^* = \hat{\mu}^*$

$$\begin{aligned}
&= \sigma^2 + \sigma^2 \left[\frac{1}{n} + \frac{(x^* - \bar{x})^2}{S_{XX}} \right] \\
&= \sigma^2 \left[1 + \frac{1}{n} + \frac{(x^* - \bar{x})^2}{S_{XX}} \right]
\end{aligned}$$

Thus,

$$y^* - \hat{y}^* \sim N \left(0, \sigma^2 \left[1 + \frac{1}{n} + \frac{(x^* - \bar{x})^2}{S_{XX}} \right] \right)$$

So then we have that

$$y^* \sim N \left(\hat{y}^*, \sigma^2 \left[1 + \frac{1}{n} + \frac{(x^* - \bar{x})^2}{S_{XX}} \right] \right)$$

Estimating the Parameters for the WLS Model

Note that $WRSS = \sum_{i=1}^n w_i (y_i - \beta_0 - \beta_1 x_i)^2$

Let's start by taking the derivative with respect to β_0

$$\begin{aligned}\frac{\partial WRSS}{\partial \beta_0} &= -2 \sum w_i (y_i - \beta_0 - \beta_1 x_i) \\ &\Rightarrow \sum w_i (y_i - \beta_0 - \beta_1 x_i) = 0 \\ &\Rightarrow \sum w_i y_i = \beta_0 \sum w_i + \beta_1 \sum w_i x_i \\ &\text{Multiply by } \sum w_i x_i \\ &\Rightarrow \sum w_i x_i \sum w_i y_i = \beta_0 \left(\sum w_i x_i \sum w_i \right) + \beta_1 \left(\sum w_i x_i \right)^2\end{aligned}$$

Now let's take the derivative with respect to β_1

$$\begin{aligned}\frac{\partial WRSS}{\partial \beta_1} &= -2 \sum w_i x_i (y_i - \beta_0 - \beta_1 x_i) \\ &\Rightarrow \sum w_i x_i (y_i - \beta_0 - \beta_1 x_i) = 0 \\ &\Rightarrow \sum w_i x_i y_i = \beta_0 \sum w_i x_i + \beta_1 \sum w_i x_i^2 \\ &\text{Multiply by } \sum w_i \\ &\Rightarrow \sum w_i \sum w_i x_i y_i = \beta_0 \left(\sum w_i \sum w_i x_i \right) + \beta_1 \left(\sum w_i \sum w_i x_i^2 \right)\end{aligned}$$

Now let's take $\frac{\partial WRSS}{\partial \beta_0} - \frac{\partial WRSS}{\partial \beta_1}$

$$\begin{aligned}\frac{\partial WRSS}{\partial \beta_0} - \frac{\partial WRSS}{\partial \beta_1} &\Rightarrow \sum w_i x_i \sum w_i y_i - \sum w_i \sum w_i x_i y_i = \beta_1 \left(\sum w_i x_i \right)^2 - \beta_1 \left(\sum w_i \sum w_i x_i^2 \right) \\ &\Rightarrow \sum w_i x_i \sum w_i y_i - \sum w_i \sum w_i x_i y_i = \beta_1 \left[\left(\sum w_i x_i \right)^2 - \sum w_i \sum w_i x_i^2 \right] \\ &\Rightarrow \beta_1 = \frac{(\sum w_i x_i \sum w_i y_i - \sum w_i \sum w_i x_i y_i) (-1)}{\left[\sum w_i \sum w_i x_i^2 - (\sum w_i x_i)^2 \right] (-1)} \\ &\Rightarrow \hat{\beta}_1 = \frac{\sum w_i (x_i - \bar{x}_w) (y_i - \bar{y}_w)}{\sum w_i (x_i - \bar{x}_w)^2} \quad \text{where } \bar{x}_w = \frac{\sum w_i x_i}{\sum w_i} \text{ and } \bar{y}_w = \frac{\sum w_i y_i}{\sum w_i}\end{aligned}$$

Finally, it follows that

$$\hat{\beta}_0 = \bar{y}_w - \hat{\beta}_1 \bar{x}_w$$